THARINDU KOTHALAWALA

Data Science Undergraduate | Machine Learning & AI Enthusiast | Mechatronics Engineer

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GitHub: github.com/nvtharindukothalawala-tech | DagsHub: dagshub.com/nv.tharindukothalawala



PROFESSIONAL SUMMARY

Data Science undergraduate at Sri Lanka Technology Campus (SLTC) with hands-on experience in machine learning, data engineering, and AI agent development, and an NVQ 5 background in Mechatronics, Robotics and Automation Engineering. Skilled in Python, Pandas, Scikit-learn, and TensorFlow, with practical project experience covering exploratory data analysis, predictive modeling, explainable AI, business intelligence, and big data pipelines. Completed the Machine Learning Specialization by DeepLearning.AI and Stanford University. Seeking a Data Science or Machine Learning role to apply strong analytical, programming, and problem-solving skills to real-world data-driven systems.

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, TypeScript, Dart, C++

Machine Learning & Data Science: Machine Learning, Deep Learning, TensorFlow, Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Feature Engineering, Model Evaluation, Regression, Classification, Exploratory Data Analysis (EDA), Statistical Analysis, Time-Series Analysis, Data Cleaning, Data Preprocessing

Artificial Intelligence: AI Agents, Large Language Models (LLMs), Prompt Engineering, LangGraph, OpenAI API, Decision Systems, Process Automation

Big Data & Databases: PySpark, MongoDB, PostgreSQL, SQLite, Firebase, Data Pipelines, RFM Analysis, Market Basket Analysis

Business Intelligence & Visualization: Power BI, Matplotlib, Jupyter Notebook, KPI Reporting, Metric Reconciliation, Executive Presentations

Backend Development & APIs: FastAPI, REST APIs, SQLAlchemy, Authentication, CRUD Operations

Tools & Platforms: Git, GitHub, GitHub Actions, Docker, DagsHub, VS Code, Streamlit

WORK EXPERIENCE

Student Intern, LAUGFS Corporation (Rubber) Ltd

Horana, Western Province, Sri Lanka | January 2023 - July 2023

- Gained hands-on industrial and technical exposure within a manufacturing environment as part of NVQ practical training.

Laboratory Assistant, Lanka Walltiles PLC (LANKATILES)

Colombo District, Western Province, Sri Lanka | January 2018 - June 2018

- Supported laboratory testing and quality-control processes on a part-time, on-site basis.

PROJECTS

Aladdin - AI Forex Trading Assistant (Individual Project, Ongoing)

Python, FastAPI, SQLAlchemy, AI Agents, Docker, GitHub Actions

GitHub: github.com/nvtharindukothalawala-tech/Aladdin-forex-ai

- Built a multi-agent AI assistant for forex market analysis, risk management, trade planning, journaling, and explainable AI coaching.
- Designed a FastAPI and SQLAlchemy backend with CI/CD via GitHub Actions and containerized deployment using Docker.

SmartCare Hospital AI (Group Project, Ongoing)

Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit

GitHub: github.com/nvtharindukothalawala-tech/smartcare-ai-xai-project

- Developing a machine learning pipeline to predict patient disease risk, including data preprocessing and model evaluation.
- Implementing explainable AI using SHAP and an interactive Streamlit application for real-time predictions.

FinSight Lanka - Customer Analytics, Dialog Finance Analytics Trainee Assessment (Individual Project, Completed)

Python, Pandas, Jupyter Notebook, Power BI

GitHub: github.com/nvtharindukothalawala-tech/FinSight-Customer-Analytics

- Analyzed 500 anonymized customers across savings behavior, loan portfolio, credit risk, digital engagement, and five monthly KPIs.
- Built a data cleaning notebook, independently reconciled metrics against a Power BI dashboard, and delivered a 10-slide executive deck.

Scalable Retail Analytics Pipeline (Group Project, Completed)

Python, PySpark, MongoDB, Big Data, Jupyter Notebook

GitHub: github.com/nvtharindukothalawala-tech/Scalable-Retail-Analytics-Pipeline

- Built an end-to-end big data pipeline processing over 541,000 retail transactions using Bronze, Silver, and Gold data layers.
- Performed RFM analysis and market basket analysis, storing curated results in MongoDB for downstream analytics.

eBay Sales Business Intelligence (Individual Project, Completed)
Python, Pandas, Scikit-learn, Jupyter Notebook, Power BI
GitHub: github.com/nvtharindukothalawala-tech/ebay-sales-business-intelligence
- Analyzed real eBay sales data through data cleaning, exploratory data analysis, and time-series analysis.
- Built machine learning models and an interactive Power BI dashboard to surface sales insights.

Lanka Smart Mart (Group Project, Completed)
Flutter, Dart, Firebase, SQLite, Google ML Kit
GitHub: github.com/nvtharindukothalawala-tech/Lanka-Smart-Mart
- Developed a full-stack e-commerce mobile application with Firebase services, offline SQLite storage, and authentication.
- Integrated AI-powered product recognition using Google ML Kit and built inventory management features.

QuizServer and Student Management App (Completed)
Electron, JavaScript, Java, Android, SQLite
GitHub: github.com/nvtharindukothalawala-tech/QuizServer | github.com/nvtharindukothalawala-tech/StudentManagementApp
- QuizServer: Electron desktop application for running quizzes over a local network, accessible via browser or QR code.
- Student Management App: Android application for managing student records with full create, read, update, and delete (CRUD) operations.

EDUCATION

BSc (Hons) in Data Science, Information Technology
Sri Lanka Technology Campus (SLTC) | October 2023 - February 2027 (Currently Studying)

NVQ 5 - Mechatronics, Robotics and Automation Engineering
Sri Lanka College of Technology, Maradana | April 2020 - July 2023 | Grade: Finalist

Diploma of Education - English Language and Literature
Aquinas College of Higher Studies | April 2018 - August 2018

Secondary Education
Bope Rajasinghe National School (BRNS) | January 2005 - December 2018

CERTIFICATIONS

- Machine Learning Specialization - DeepLearning.AI and Stanford University (July 2026)
- Advanced Learning Algorithms - DeepLearning.AI and Stanford University (July 2026)
- Unsupervised Learning, Recommenders, Reinforcement Learning - DeepLearning.AI and Stanford University (July 2026)
- Supervised Machine Learning: Regression and Classification - DeepLearning.AI and Stanford University (July 2026)

LEADERSHIP & ACTIVITIES

- **Head Prefect**, School Prefects' Board, Bope Rajasinghe National School: led a team of prefects, coordinating discipline, events, and student welfare initiatives.
- **Member**, Leo Club of Sri Lanka Technology Campus (August 2024 - Present): community service and volunteering initiatives.
- **Member**, AIESEC (March 2024 - Present): youth leadership and cross-cultural global volunteering programs.
- **Member**, LEO Club of Padukka (January 2022 - Present): local community service projects and volunteering.

REFERENCES

Dr. Chameera De Silva, PhD (UniMelb)
Head of the Department - Data Science / Lecturer
FOCIT - Faculty Industry Placement Coordinator
Member of Ethics Review Committee
Member of Technical Review Committee
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